

Minimally Invasive Neural Interface for Long-Term Brain Monitoring and Rehabilitation

(Executive Summary)

This Project presents the design, prototyping, and validation plan for a **minimally invasive neural interface** that combines the high signal fidelity of electrocorticography (ECoG) with the safety and usability of noninvasive systems. The solution is a screw-sized, skull-anchored **bolt** housing integrated microelectronics (amplifier, ADC, power management, and wireless telemetry). After a **one-time minor surgical placement** with the skin closed over the device, the system enables wireless, low-profile cortical signal readout for multi-day monitoring and intermittent use. Primary application is epilepsy monitoring; secondary applications include motor prosthesis control, rehabilitation for paralysis/amputation, and future bidirectional stimulation platforms.

Background & Problem Statement

Noninvasive EEG is safe and accessible but suffers from poor spatial resolution and attenuated signal amplitudes after passing through scalp, skull, and meninges. Invasive ECoG and intracortical arrays provide superior spatial and temporal fidelity but require craniotomy, expose patients to infection risk, and are not practical for long-term ambulatory monitoring. Current clinical need includes continuous or multiday cortical monitoring for epilepsy and ambulatory neural sensing for rehabilitation. There is a gap for an implantable architecture that maintains skin integrity, reduces infection risk, and provides near-cortical signal quality.

Proposed Solution — System Overview

Form factor & surgical workflow

- A screw-sized bolt anchored into the skull; blunt atraumatic tip rests on the dura mater without penetration.
- One-time minor surgical placement through a small bone recess; the scalp is closed so no percutaneous hardware remains.
- External receiver/charger (flexible form factor): headband, cap, or small wearable; earbud-sized device is optional, not mandatory.

Core electronics

- Low-noise analog front-end (AFE) with adjustable gain and active filtering.
- ADC with programmable sample rates and resolution suited for cortical local field potentials (LFPs) and event detection (~250–2,000 Hz sampling range depending on mode).
- Power management: small rechargeable cell as PoC; options for inductive recharge or NFC/interrogation to minimize battery capacity.
- Wireless telemetry modes: continuous low-duty streaming (BLE/low-power Wi-Fi) and NFC-style on-demand readouts to conserve energy.

Operational modes

- **Continuous monitoring** for high-risk or inpatient scenarios.
- **Event-driven capture** where the device logs or streams only on detection/trigger to preserve power.
- **Scheduled or on-demand download** via an external reader for routine ambulatory use.

Technical Design Considerations

Signal acquisition & front-end

- Input signals are μV -scale; AFE must have ultra-low input-referred noise ($< \text{nV}/\sqrt{\text{Hz}}$ targets), high CMRR, and differential recording capability where possible.
- Active electrode impedance monitoring and calibration routines to detect coupling degradation.

Electrode & mechanical interface

- Candidate electrode materials: platinum/iridium (Pt/Ir), iridium oxide (IrOx), or coated electrodes (PEDOT, titanium nitride) for low impedance and charge stability.
- Mechanical coupling must ensure gentle contact force on dura, minimize micro-motion artifacts, and tolerate skull remodeling over time.

Power, thermal, and safety

- Conservative thermal budget (max local $\Delta T < 1^\circ\text{C}$ over baseline during continuous operation); firmware enforced duty cycles and thermal shutdowns.
- Protective circuits for safe charging, overcurrent/short protection, and chemical safety for cell choice.

RF, link budget, and regulatory

- RF propagation through skull/skin modeled for candidate bands; optimize transmit power, duty cycle, and modulation for reliability and compliance.
- Early regulatory consultation (local medical device agency and spectrum authorities) to select medically suitable bands and preempt classification pathways.

Mechanical impact & longevity

- Housing designed to decouple external impact loads from the dura (compliant intermediate structure, crush stops).
- Materials and coatings chosen for long-term biostability and minimal fibrotic reaction.

Evaluation & Validation Plan

Bench & Phantom Testing

- Construct tissue phantoms replicating skull, dura, and cortical conductivities for signal capture and RF transmission testing.
- Metrics: SNR across frequency bands (delta–gamma), input impedance over time, thermal profile under continuous and burst transmissions, packet loss/latency for wireless links.

Iterative Prototyping

- Build initial 3D-printed mechanical housings and breadboard electronics for rapid test cycles.
- Validate amplifier performance with microvolt test signals and simulated neural waveforms.

Preclinical Steps

- If bench results are promising, plan for *ex vivo* cadaveric skull tests and small animal studies to validate placement, tissue response, and RF path behavior.
- Define ethical and regulatory milestones before any in-vivo human work.

Success Criteria

- Demonstrate SNR and signal fidelity comparable to subdural surface recordings for target frequency bands.
- Thermal rise within safe limits under worst-case operation.
- Wireless throughput and latency adequate for intended monitoring mode.
- Stable electrode impedance over the PoC test duration.

Risks, Mitigations & Open Questions

- **Signal degradation over chronic timescales:** mitigate via material selection, encapsulation, and firmware calibration/checks; design modular electronics for future upgrades.
- **Thermal & power failure modes:** extensive thermal simulation, hardware watchdogs, and conservative duty cycles.
- **Biological response & infection:** maintain intact skin, use proven implantable materials, and follow surgical asepsis protocols. Early preclinical histology will guide material choices.
- **Regulatory classification & RF issues:** engage regulatory/clinical advisors early to map testing and submission requirements.
- **Surgical adoption & ergonomics:** co-design surgical steps with neurosurgeons to ensure reproducibility and safety.

Expected Impact & Applications

- **Clinical monitoring:** multi-day, ambulatory cortical recordings for epilepsy localization and postoperative surveillance without percutaneous hardware.
- **Neurorehabilitation:** control signals for prosthetics, exoskeletons, and assistive devices by capturing motor cortical LFP signatures.
- **Therapeutic extension:** a future pathway for inductive stimulation and closed-loop neuromodulation for seizure prevention or motor recovery.

Conclusion

This work proposes a pragmatic, risk-aware path toward a minimally invasive cortical sensing platform that maintains skin integrity after implant placement and provides near-ECoG fidelity without chronic percutaneous leads. The planned PoC, validation, and regulatory strategy are designed to produce credible technical evidence supporting clinical translation and broader rehabilitation applications.